

KHANTY-MANSIYSK, Russian Federation

WMO: 239330

Lat: 61.0282N

Lon: 69.1179E

Elev: 23

StdP: 101.05

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.9	-35.1	-40.9	0.1	-37.7	-38.1	0.1	-35.0	7.9	-13.5	7.3	-13.1	1.1	100	0.635

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	28.1	19.4	26.2	18.8	24.4	18.0	20.9	25.9	19.8	24.4	18.8	23.4	3.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	23.6	18.0	13.0	22.5	16.9	12.1	21.5	60.6	26.0	57.0	24.6	53.5	23.6	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 7.4	(b) 6.5	(c) 5.8	(e) -40.4	(f) 31.2	(g) 4.1	(h) 1.8	(i) -43.4	(j) 32.5	(k) -45.8	(l) 33.6	(m) -48.1	(n) 34.6	(o) -51.2	(p) 35.9		
(a) 7.4	(b) 6.5	(c) 5.8	(e) -40.3	(f) 23.1	(g) 4.2	(h) 1.2	(i) -43.3	(j) 24.0	(k) -45.8	(l) 24.6	(m) -48.2	(n) 25.3	(o) -51.2	(p) 26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.4	-19.6	-17.1	-7.7	-0.2	8.0	15.4	18.1	14.6	8.3	0.7	-10.4	-16.5
	DBStd	14.54	8.94	8.45	6.50	6.09	5.81	5.04	4.08	4.06	4.34	5.19	8.27	9.03
	HDD10.0	4469	918	758	549	309	111	12	1	6	84	290	611	821
	HDD18.3	6947	1177	992	807	557	325	114	57	130	302	547	861	1079
	CDD10.0	657	0	0	0	2	48	175	251	148	32	2	0	0
	CDD18.3	93	0	0	0	0	3	27	49	13	1	0	0	0
	CDH23.3	637	0	0	0	0	32	189	328	86	2	0	0	0
	CDH26.7	127	0	0	0	0	5	40	66	16	0	0	0	0
Wind	WSAvg	2.6	2.4	2.4	3.0	3.0	2.8	2.6	2.1	2.1	2.4	2.8	2.8	2.6
Precipitation	PrecAvg	538	28	22	24	30	43	63	76	73	58	51	40	32
	PrecMax	757	68	51	81	64	110	169	233	214	157	114	79	57
	PrecMin	352	4	4	0	2	5	19	14	9	14	16	10	4
	PrecStd	93	13	11	15	15	24	29	50	35	31	24	17	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.3	1.1	6.1	17.9	27.0	30.1	30.5	28.9	23.1	14.9	5.0	0.9
		MCWB	-0.2	0.0	2.5	11.7	15.5	19.4	21.1	21.1	16.7	10.6	4.2	0.3
	2%	DB	-2.7	-1.2	3.5	12.4	23.3	27.7	28.5	25.8	19.8	12.0	2.6	-0.8
		MCWB	-3.1	-2.0	0.8	7.5	14.2	18.5	20.0	19.3	14.7	8.9	1.9	-1.4
	5%	DB	-5.1	-3.7	1.9	10.0	20.5	25.6	26.8	23.2	17.0	9.9	1.2	-2.6
		MCWB	-5.5	-4.3	0.1	5.7	13.2	18.0	19.5	18.0	13.0	7.8	0.4	-3.1
	10%	DB	-7.3	-5.7	0.5	7.8	17.5	23.1	25.1	21.2	14.8	7.7	0.0	-4.9
		MCWB	-7.8	-6.5	-1.0	4.3	11.5	17.0	18.9	16.9	11.9	6.2	-0.6	-5.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.1	0.3	2.6	11.8	17.2	21.5	22.7	21.7	17.5	11.2	4.1	0.2
		MCDB	0.3	1.0	5.1	17.1	23.8	26.4	27.6	27.0	22.2	13.5	4.7	0.7
	2%	WB	-3.1	-2.2	1.2	8.5	15.2	19.7	21.3	20.0	15.1	9.4	2.0	-1.3
		MCDB	-2.7	-1.4	3.0	11.7	22.0	25.0	26.4	24.3	18.9	11.7	2.4	-0.8
	5%	WB	-5.6	-4.3	0.2	6.1	13.6	18.7	20.4	18.6	13.6	7.8	0.6	-3.1
		MCDB	-5.3	-3.7	1.6	9.5	19.5	24.2	25.1	22.5	16.5	9.5	1.0	-2.6
	10%	WB	-7.8	-6.3	-0.8	4.4	11.9	17.5	19.5	17.4	12.0	6.1	-0.7	-5.4
		MCDB	-7.3	-5.6	0.5	7.4	17.5	22.6	24.0	20.7	14.6	7.5	-0.1	-4.9
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.4	8.3	8.5	9.3	8.3	8.3	7.8	6.9	5.2	5.9	7.2
		MCDBR	7.6	6.5	7.8	10.4	12.7	11.3	10.9	10.8	9.9	7.8	4.8	7.6
		MCWBR	7.6	6.3	6.2	6.8	6.7	5.5	4.8	5.6	6.0	5.2	4.7	7.4
	5% WB	MCDBR	7.8	6.5	7.0	10.0	11.9	10.3	9.7	9.9	9.3	7.3	5.0	7.8
		MCWBR	7.8	6.4	5.9	7.0	6.7	5.6	4.8	5.6	6.1	5.3	5.0	7.7
Clear-Sky Solar Irradiance	taub		0.257	0.302	0.354	0.382	0.395	0.407	0.421	0.415	0.358	0.327	0.273	0.241
	taud		1.975	2.010	1.957	2.058	2.271	2.330	2.282	2.257	2.380	2.272	2.043	1.914
	Ebn at Noon		432	640	728	797	825	821	796	764	752	628	409	273
	Edh at Noon		44	82	122	134	118	114	117	110	82	64	41	29
All-Sky Solar Radiation	RadAvg		0.40	1.28	2.83	4.34	5.03	5.36	5.18	3.67	2.26	1.00	0.38	0.20
	RadStd		0.03	0.10	0.17	0.45	0.41	0.42	0.51	0.35	0.23	0.12	0.07	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page